higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## PULP AND PAPERMAKING TECHNOLOGY
### NQF LEVEL 3

(6050383)

**14 November 2019 (X-Paper)**
**09:00–12:00**

**This question paper consists of 6 pages.**

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**TIME: 3 HOURS**
**MARKS: 100**

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**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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## QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 Bark and plastics can strengthen the quality of paper.

1.2 The type of de-inking process used depends on the amount of ink that needs to be removed.

1.3 Sulphuric acid removes lignin from the chips.

1.4 During counter current washing, the wash water flow in the same direction as the pulp.

1.5 The laminating process hides faults such as holes, lumps and creases in and on the paper.

(5 × 1) **[5]**

## QUESTION 2

Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–G) next to the question number (2.1–2.5) in the ANSWER BOOK.

| COLUMN A | | COLUMN B | |
|---|---|---|---|
| 2.1 | Soft spongy centre of plant stems | A | opacity |
| 2.2 | Equipment in which fibre bundles are broken down into loose, stretchy fibres | B | bagasse |
| | | C | calendar |
| 2.3 | A test that shows how well paper stops light from passing through | D | brightness |
| | | E | delignification |
| 2.4 | The process of removing lignin from the pulp | F | pith |
| 2.5 | A pair of rolls used for squeezing or pressing | G | deflaker |

(5 × 1) **[5]**

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## QUESTION 3

Give ONE word or term for each of the following descriptions by choosing a word or term from the list below. Write only the word or term next to the question number (3.1–3.5) in the ANSWER BOOK.

3.1 A large hole in the ground where waste material is discarded

3.2 Activities that are performed to ensure that a process stays stable and continues to perform at the best possible level

3.3 A component in groundwood pulping that is used to keep the stone from burning and becoming damaged

3.4 A path created in the pulp by the chemical flow inside it

3.5 A shiny and very smooth paper that is used for magazines and brochures

(5 × 1) **[5]**

## QUESTION 4

4.1 The logs that come into the woodyard are turned into woodchips before being converted into pulp.

4.1.1 List SEVEN steps in sequence for turning logs into chips. (7)

4.1.2 Describe the working principles of the woodyard. (6)

4.2 Choose the correct equipment used in the process of making pulp from bagasse from the equipment given in brackets to match a description given below. Write only the correct equipment next to the question number (4.2.1–4.2.4) in the ANSWER BOOK.

| pin drum feeder; dewatering press; bagasse washer; depither |
|---|

4.2.1 Removes unwanted diet and contraries from bagasse

4.2.2 Separates fibre from pith

4.2.3 Controls the amount of bagasse fed to the process

4.2.4 Dries pith and sludge before being thrown into a landfill

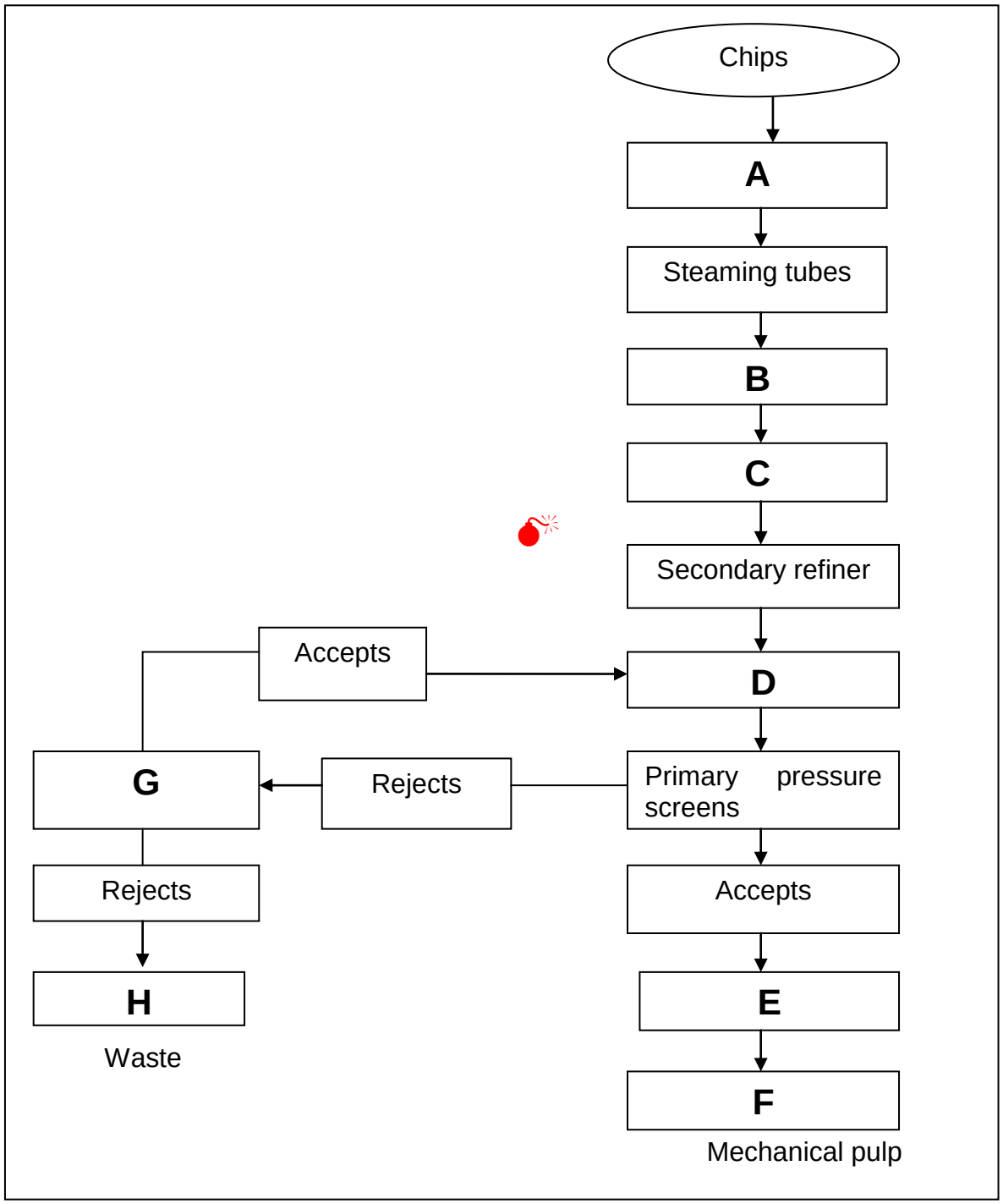
(4 × 1) (4)

**[17]**

**QUESTION 5**

5.1 List FIVE physical tests that can be done on paper produced from secondary fibre. (5)

5.2 Complete the diagram below of a thermomechanical process by naming the processes (A–H) in the ANSWER BOOK.




(8)

5.3 State FOUR points to be considered to ensure that the shower system of a mechanical pulping process functions effectively. (4)

**[17]**

## QUESTION 6

6.1 What is the main purpose of softening the chips by using steam? (2)

6.2 Why does Kraft pulping produce a lower yield than mechanical pulping? (1)

6.3 What would you do to make sure that the digester is properly filled and chips are in the right position to avoid a pyramid digester? (3)

6.4 Give FIVE reasons for large amounts of unfiberised wood in the Kraft pulping process. (5)

6.5 Name TWO types of equipments used for soda pulping process. (2)

6.6 What are the FOUR indicators of poor acid sulphite pulping performance? (4)
**[17]**

## QUESTION 7

7.1 The quality of the pulp has a direct effect on the quality of paper that will be made.

List FOUR quality requirements of bleached pulp. (4)

7.2 How would you remedy poor bleaching performance? (5)

7.3 Which instrument will you use to measure the alkalinity or acidity of the pulp? (1)

7.4 Sketch a labelled diagram of rotary vacuum washer regions. (7)
**[17]**

## QUESTION 8

8.1 Embossing is a finishing process that puts a design or image on the surface of the paper.

8.1.1 Give FOUR steps to be followed during the embossing process. (4)

8.1.2 State TWO indicators of poor embosser performance. (2)

8.2 Give TWO reasons why paper is coated. (2)

8.3 Explain the working principles of de-aerators with reference to coating equipment. (5)

8.4 Differentiate between *natural binders* and *synthetic binders.* (2 × 2) (4)
**[17]**

**TOTAL: 100**

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